

1

$$y = x^2 - 4x + 3$$

$$(2, -1) \quad x = 2 \quad y = (x-2)^2 - 1$$

$$x = -b/2a$$

2

$$y = ax^2 + bx + c \quad x \quad (1,0) \quad (5,0) \quad x = 3 \quad y = 4 \quad a$$

$$y = a(x-1)(x-5) \quad x=3 \quad 4 = a(2)(-2) \quad a = -1$$

$$a(x-1)(x-5)$$

3

$$k \quad x^2 - (k+1)x + k = 0$$

$$D = (k+1)^2 - 4k = k^2 - 2k + 1 = (k-1)^2 \quad D > 0 \quad k \neq 1$$

$$D > 0 \quad k$$

4

$$y = -x^2 + 6x - 5 \quad x$$

$$(3, 4) \quad 4$$

5

$$24 \quad x \quad A(x)$$

$$= 24 - 2x \quad A(x) = x(24 - 2x) = 24x - 2x^2 \quad x = 6 \quad A = 72$$

$$A$$

6

$$2 \quad 3 \quad y = x^2$$

$$y = (x - 2)^2 + 3 \quad (2, 3)$$

$$y - k = (x - h)^2$$

jjredick365@gmail.com